# Long-Giang Vu

giangvl.cs@gmail.com · ginlov.github.io · Google Scholar · LinkedIn · GitHub

## Research Interests

Machine learning for robot motion planning: learned planners with safety and feasibility guarantees, structured 3D scene representations for planning under partial observability, and vision-language-action models grounded in geometry and physics.

## Education

**University of California, San Diego** 2024 – 2026

*M.S. Computer Science · GPA 3.86/4.0 · VinGroup Scholarship (full tuition and living expenses)* *La Jolla, CA*

- Research advisor: Prof. Ravi Ramamoorthi. Fine-tuned the 2B-parameter Stable Virtual Camera model for high-fidelity multi-view human generation.
- Built synthetic datasets to train lighting-variant (IC-Light) and pose-variant (InfiniU) generative models; injected face embeddings to improve identity preservation.
- Coursework: Sensing & Estimation in Robotics, Mobile Manipulation, Autonomous Vehicles, Reinforcement Learning, Physics Simulation, Convex Optimization, Differentiable Programming, Computer Vision.

**Hanoi University of Science and Technology** 2018 – 2023

*B.S. Computer Science · GPA 3.76/4.0* *Hanoi, Vietnam*

- Thesis advisor: Assoc. Prof. Khoat Than. Thesis: *Improving Diffusion Model Inference by Dropout*, highest score at the Machine Learning thesis committee.
- Research on diffusion inference and generalization of over-parameterized networks (AAAI 2024 oral; *Machine Learning* journal).

## Publications

* equal contribution

1. **Long-Giang Vu***, Chengkai Yao*, Yuxin Liu, Rajath C. Aralikatti, Aryan. AnchorVLN: Geometry-Anchored Vision-Language Grounding Reasoning for Open-Vocabulary Navigation. *IROS 2026 Workshop on Bridging Perspectives in Navigation (Bridge4Navigation)*.
2. Khoat Than, Dat Phan, **Giang Vu**. Gentle Local Robustness Implies Generalization. *Machine Learning* (Springer Nature), 114(6):142, 2025.
3. **Giang Vu***, Viet Nguyen*, Tung Nguyen Thanh, Khoat Than, Toan Tran. On Inference Stability for Diffusion Models. *AAAI 2024*, **oral presentation**.

## Research and Robotics Projects

**Vision-Language Navigation on a Mobile Robot · Team Xiao Hei (team lead)** Jun – Sep 2026

*code* · *project page*

- **2nd place, real-robot track** and 4th place, simulation track, CMU Vision-Language Navigation Challenge 2026 (IROS 2026); led a five-person team.
- Built the perception stack that turns a 360° camera and LiDAR into a cumulative 3D scene graph: open-vocabulary detection (YOLO-World + SAM 2.1) lifted to 3D through the registered LiDAR scan and fused into persistent object nodes at 2 Hz, with panorama-seam merging, novelty gating and image/pose/LiDAR sensor synchronization.
- Decoupled exploration and scene capture from LLM reasoning: frontier-based exploration on an occupancy grid built from the terrain map collects the scene graph, panorama and trajectory map, and the language model answers only once the sweep completes.
- Added question-directed VLM navigation and coverage path planning, plus an offline multi-scene benchmark for perception and object grounding.
- AnchorVLN (co-first author): designed the scene graph and the robot–VLM communication protocol, and built the `expand_vocab` and `verify_present` perception skills and the `explore` and `orbit` motion skills of its 10-tool interface, where the VLM reasons over semantics and LiDAR geometry supplies all metric quantities. On the 15-scene development set it reaches 64.4% instruction-following success vs. 51.1% without the platform model, and 22% object-grounding hits vs. 0% for direct VLM estimation. The same interface runs on a humanoid by swapping only the controller.

**Learned Motion Planning for Autonomous Driving with GRPO** page

- Extended a Wayformer planner with a temporal Gaussian decoder and fine-tuned it with Group Relative Policy Optimization, using a collision-penalizing reward over sampled trajectory proposals to suppress unsafe expert-like modes in long-tail scenarios.

**Robotics Coursework Projects, UCSD** 2025 – 2026

- *State estimation and SLAM (ECE 276A):* LiDAR SLAM with ICP and GTSAM pose-graph optimization cut trajectory error by 76% and 72% on two datasets; built EKF visual-inertial SLAM on SE(3).
- *Mobile manipulation (MAE 204):* feedforward plus PI task-space control for pick-and-place on a KUKA youBot.
- *Autonomous car (MAE 148):* built a ROS 2 car that detects a stranded vehicle from a roadside camera, then approaches and parks using detection, LiDAR ranging and PID control.

## Industry Experience

**Machine Learning Engineer, Woven by Toyota** Apr 2026 – present

- Integrated a new driving-data source of tens of thousands of hours into training to improve the generalization of a vision-only end-to-end driving model.
- Built domain transfer from the in-house 11-camera setup to the 5-camera setup of market vehicles, opening fleet data as a source of long-tail scenes for training.
- Built shadow-mode evaluation that captures bad driving behavior during real-vehicle autonomous tests using end-to-end metrics.

**Software Engineer Intern, Woven by Toyota** Jun – Aug 2025 · California, US

- Designed metrics that quantify training-data quality for the motion planning stage of an autonomous driving system (manager: Dr. Alborz Alavian).
- Worked across the autonomy stack: map representation, safety optimization framework and vehicle operating system.

**Data Scientist, Techcom Securities** May – Aug 2024 · Hanoi, Vietnam

- Cut experiment and campaign launch cycles from 1–2 weeks to 10–20 minutes by consolidating scattered data tasks into one AutoML platform.

**Data Application Engineer, Viettel** Jun 2023 – Apr 2024 · Hanoi, Vietnam

- Architected the Viettel Machine Learning Platform and Recommendation Platform covering the end-to-end data science workflow.

**AI Engineer, TopCV** Jul 2021 – Apr 2023 · Hanoi, Vietnam

- Built a re-ranking algorithm that raised conversion by 2% in high-traffic sections, plus an internal NLP toolkit and job-spam detection.

## Honors and Awards

- 2nd place, real-robot track, CMU Vision-Language Navigation Challenge, IROS 2026.
- Oral presentation, AAAI 2024.
- VinGroup Scholarship for graduate study at UCSD, full tuition and living expenses, 2024.
- HUST Outstanding Student Scholarship, top 1%, Fall 2020 and Fall 2021.

## Skills

- **Robotics:** ROS 2, occupancy mapping and frontier exploration, SLAM (ICP, GTSAM, EKF), LiDAR–camera fusion, PID and task-space control.
- **Machine learning:** PyTorch, diffusion models, reinforcement learning (GRPO), open-vocabulary perception (YOLO-World, SAM 2), VLM tool use.
- **Engineering:** Python, Docker, ML platform design, large-scale fine-tuning.